

CS 111

Foundations of Program Design

Week 6, Wednesday Lecture

Roadmap

- Week 8: Guest lectures
- Week 9 - 11: Write interactive Java programs using Processing
- Week 12 - 15: Write our own Java classes
- Week 16: Final exam review

Guest lectures next week



- Monday: Jeff Gates
 - Co-founder at Temple Gates Games; previously Maxis, Blizzard, Popcap.
 - **The craft of programming** - what's the tradeoff of using Java versus other languages?

Guest lectures next week



- Wednesday: EJ Jung
 - Assistant Professor of Computer Science at USF
 - **Cybersecurity clinic** - you'll build a password cracker, combining a dictionary with substitution patterns.

Lecture + lab today

- **Review:** Assigning a new value to a variable
- Assigning a new array reference to a variable

Lab: Chi-squared test - how different are these two curves?

Variables are references to values

Primitive values are just bytes:

```
int x = 0;  
// on the left hand side of `=` , x is a reference  
x = 42;  
  
// on the right hand side, x is a value  
int y = x;  
y = 12; // x still equals 42
```

Variables are references to values

The index operator `[]` transforms the array reference into the type of value it contains:

```
double[] argument = new int[]{1.0, 2.0, 3.0};  
int i = 0;  
double x = argument[i];  
// now x == 1.0
```

Variables are references to values

On the left hand side of `=` assignment, array values (and object values) are *references*:

```
int[] argument = new int[]{1,2,3};  
int[] result = argument;  
  
result[0] = 9;  
// now argument[0] = 9
```

Lab assignment

- Given two probability distributions (as `double[]` arrays), take a sum of squared differences ('chi squared test')
- Work on your Project 1 brief (see `Week 6/Project1.md`)

Whiteboard

```
public static double distance(double[] observed, double[] expected) {  
  
    double sum = 0.0;  
    for (int i = 0; i < observed.length; i++) {  
  
        double diff = observed[i] - expected[i];  
        sum += (diff * diff) / expected[i];  
    }  
    return sum;  
}
```